Tarea lección 3.9

1292

280

3)

$A = l^2$

$A = l^2$
$16 = l^2$
$\sqrt{16} = l$
$4 = l$

$\frac{dl}{dt} = 6cm/s$

$\frac{dA}{dt} = 2l\left(\frac{dl}{dt}\right)$

$\frac{dA}{dt} = ?$ $A = 16cm^2$

$\frac{dA}{dt} = 2l \cdot \frac{dl}{dt}$

$\frac{dA}{dt} = 2(4) \cdot 6 = \underline{48cm^2/s}$

5)

$V = \pi r^2 h$
$V = \pi (5)^2 h$
$V = 25\pi h$

$r = 5m$

$\frac{dV}{dt} = 3m^3/min$

$\frac{dV}{dt} = 25\pi \frac{dh}{dt}$

$3 = 25\pi \frac{dh}{dt}$

$\frac{3}{25\pi} = \frac{dh}{dt} = \underline{\frac{3}{25\pi} m/min}$

7) $A = 4\pi r^2$

$\frac{dr}{dt} = 2cm/min$

$\frac{dA}{dt} = 4\pi 2r \cdot \frac{dr}{dt}$

$\frac{dA}{dt} = ?$ $r = 8cm$

$\frac{dA}{dt} = 4\pi 2(8)(2) = \underline{128\pi cm^2/m}$

71) $x^2 + y^2 + z^2 = 9$ $\quad (x, y, z) = (2, 2, 1)$

$\frac{dx}{dt} = 5 \qquad \frac{dy}{dt} = 4$

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} + 2z\frac{dz}{dt} = 0$

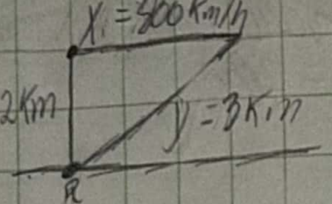
$2(2)(5) + 2(2)(4) + 2(1)\frac{dz}{dt} = 0$

$20 + 16 + 2\,dz/dt = 0$

$36 + 2\,dz/dt = 0$

$\frac{dz}{dt} = \frac{-36}{2} = \underline{-18}$

73)



$y^2 = x^2 + 2^2$

$2y\frac{dy}{dt} = 2x\frac{dx}{dt} + 0$

$y\frac{dy}{dt} = x\frac{dx}{dt}$

$3\frac{dy}{dt} = \sqrt{5}(800)$

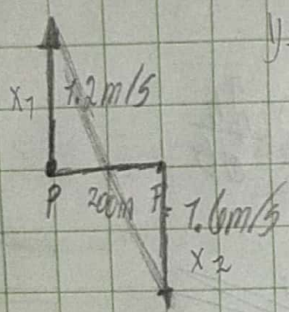
$\frac{dy}{dt} = \underline{\frac{800\sqrt{5}}{3}}$ Km/h

$x^2 + 2^2 = 3^2$

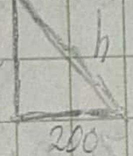
$x^2 + 4 = 9$

$x_1 = \sqrt{5}$

79)



$y = x_1 + x_2$



$h^2 = y^2 + 200^2$

$2h\frac{dh}{dt} = 2y\frac{dy}{dt}$

$dh/dt = y/h \; dy/dt$

$x_1 = 1.2$ m/s $\cdot$ 20 min $\cdot \frac{60s}{1m}$

$x_1 = 1440$ m

$x_2 = 1.6$ m/s $\cdot$ 15 min $\cdot \frac{60s}{1m}$

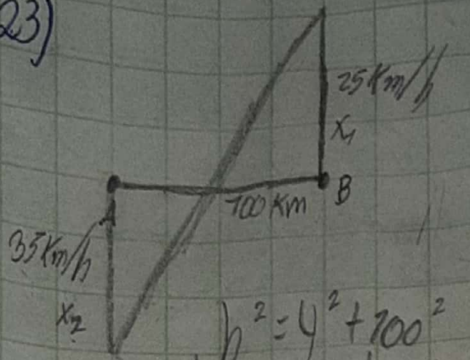
$x_2 = 1440$ m

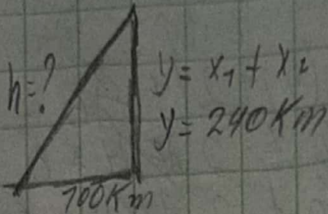
$y = 1440 + 1440 = 2880$ m

$h = \sqrt{2880^2 + 200^2} = 2886.94$

$\frac{dh}{dt} = \frac{2880}{2886.94} \cdot 2.8 = \underline{2.79 \text{ m/s}}$

23)





$x_1 = 25 * 4h = 100 Km$

$x_2 = 35 * 4h = 140 Km$

$y = 100 + 140 = 240 Km$

$h = \sqrt{240^2 + 100^2}$

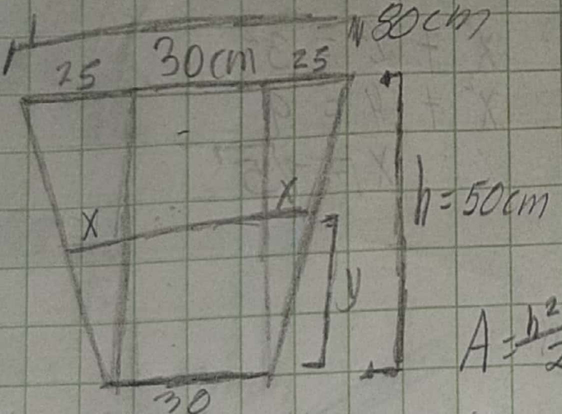
$h = 260$

$h^2 = y^2 + 100^2$

$2h \frac{dh}{dt} = 2y \frac{dy}{dt} + 0$

$\frac{dh}{dt} = \frac{y}{h} \cdot \frac{dy}{dt} = \frac{dh}{dt} = \frac{240}{260} \cdot 60 = \frac{720}{13}$ Km/h

27)



$\frac{25}{x} = \frac{50}{y}$

$A = \left[\frac{(h+30)+30}{2}\right]h = \frac{h^2}{2} + 30h$

$A = \frac{h^2}{2} + 30h$

$\frac{dA}{dt} = (h+30)\frac{dh}{dt}$

$h = 2x$

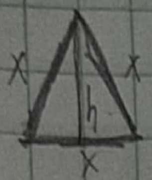
$2x + 30 = h + 30$

$\frac{dV}{dt} = l(h+30)\frac{dh}{dt}$

$0.2 \cdot 10^6 = 1,000(30+30)\frac{dh}{dt}$

$\frac{dh}{dt} = 10/3$ cm/min

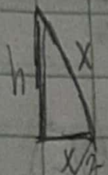
31)



$\frac{dx}{dt} = 10\,cm/min$

$A = \frac{bh}{2}$

$\frac{dA}{dt} = ?$ $\quad l = 30\,cm$



$x^2 = h^2 + \left(\frac{x}{2}\right)^2$

$h = \sqrt{x^2 - x^2/4}$

$h = \sqrt{\frac{4x^2 - x^2}{4}}$

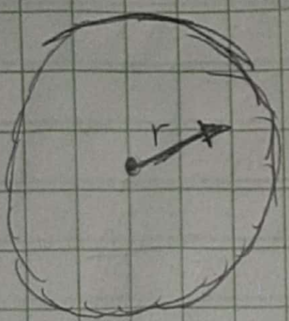
$h = \frac{x\sqrt{3}}{2}$

$A = \frac{x\left(\frac{x\sqrt{3}}{2}\right)}{2} = \frac{\sqrt{3}\,x^2}{4}$

$\frac{dA}{dt} = \frac{\sqrt{3}}{2}\,x\,\frac{dx}{dt}$

$\frac{dA}{dt} = \frac{\sqrt{3}}{2}(30)(10) = \underline{150\sqrt{3}\,cm^2/m}$

35)

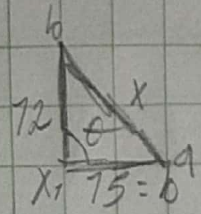


$A = \pi r^2$

$\frac{dA_c}{dt} = \pi\left(2r \cdot \frac{dr}{dt}\right)$

$\frac{dA}{dt} = \underline{\pi 2r^2\,cm^2/h}$

41)



$\frac{d\theta}{dt} = \frac{2°}{min}$ $\quad \theta = 60°$ $\quad \frac{dx}{dt} = ?$

$x^2 = a^2 + b^2 - 2ab\cos x_1$

$x^2 = 12^2 + 15^2 - 2(12)(15)\cos x_1$

$x^2 = 369 - 360\cos x_1$

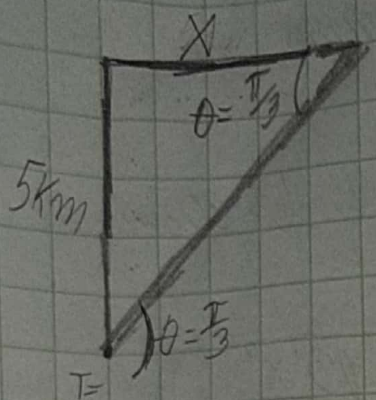
$2x\frac{dx}{dt} = -360\left(-\operatorname{sen} x_1 \frac{d\theta}{dt}\right)$

$x = \sqrt{12^2 - 2(15)(15)\cos 60}$

$x = 3\sqrt{21}$

$\frac{dx}{dt} = \frac{360\operatorname{sen} x_1 \frac{d\theta}{dt}}{2x} = \frac{360\operatorname{sen}60°\,(\pi/90)}{2(3\sqrt{21})} = 0.4$

45)



$\frac{d\theta}{dt} = \frac{\pi}{6}$ rad/min

$\tan\theta = 5/x$

$x = 5/\tan\theta = 5/\tan(\pi/3) = \frac{5\sqrt{3}}{3}$

$\frac{dx}{dt} = \frac{-x\sec^2\theta\frac{d\theta}{dt}}{\tan\theta} = \frac{-\frac{5\sqrt{3}}{3}\cdot\frac{1}{\cos^2\pi/3}\left(\frac{\pi}{6}\right)}{\tan(\pi/6)}$

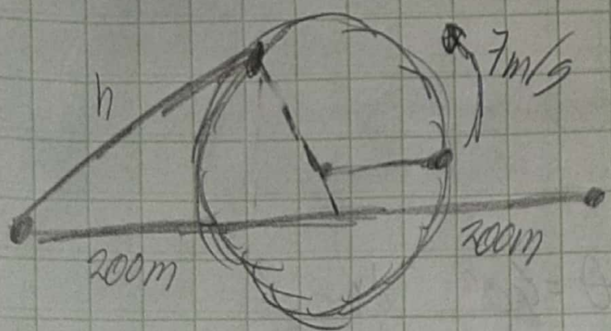
$\tan\theta = \frac{5}{x}$

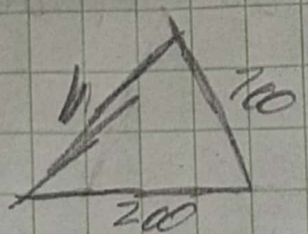
$x\tan\theta = 5$

$x\sec^2\theta\frac{d\theta}{dt} + \tan\theta\frac{dx}{dt} = 0$

$\frac{dx}{dt} = \frac{10\pi}{3}$ km/min

49)





dh/dt

$h^2 = 200^2 + 200^2 - 2(200)(200)\cos\theta$

$h^2 = 50{,}000 - 40{,}000\cos\theta$

$s = \theta r$

$s = \theta \cdot 200$

$\frac{ds}{dt} = 200\frac{d\theta}{dt}$

$\frac{7}{200} = \frac{d\theta}{dt}$